

Practical 8: LINQ

Aim:

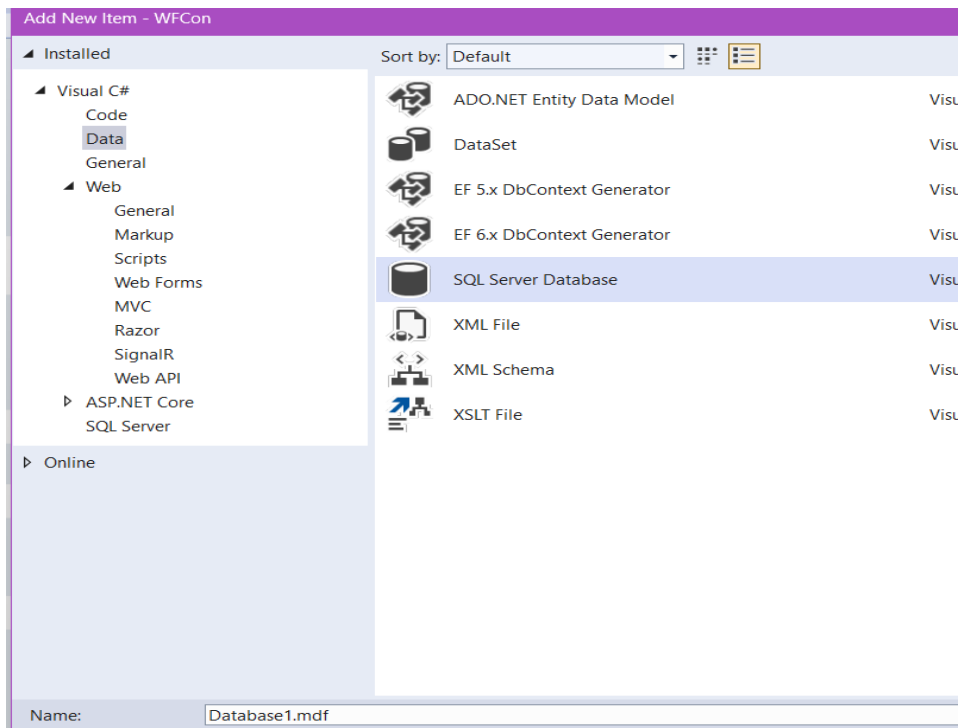
To build a website to demonstrate the working of Entity Framework in .NET.

Step 1: Create Project

Open Visual Studio → Create New Project → ASP.NET Web Application (.NET Framework) → Select Web Forms.

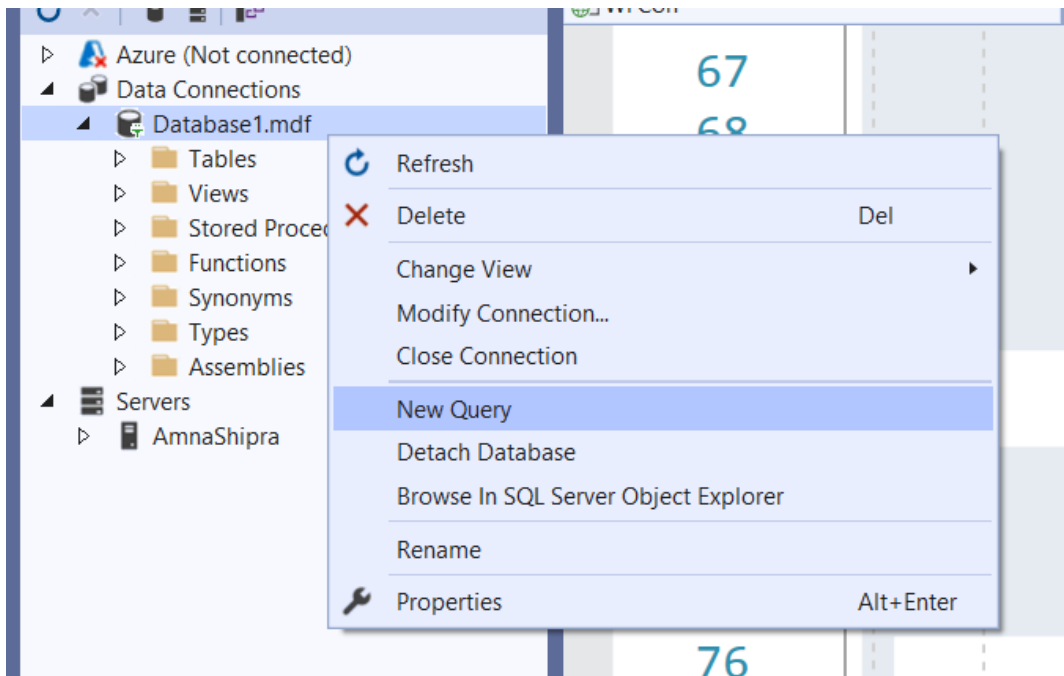
Step 2: Add Database

1. Right-click Project → **Add** → **New Item**
2. Select **SQL Server Database**
3. Name → Database1.mdf



Step 3: Create Table

Right Click on Database → New Query



Run this query:

```
CREATE TABLE Student
(
    Id INT PRIMARY KEY,
    Name VARCHAR(100),
    Age INT,
    Div VARCHAR(50),
    DreamJob VARCHAR(100)
);
```

Step 4: Add Entity Framework Model

Right Click Project → Add → New Item → ADO.NET Entity Data Model → EF Designer from Database → Select Database → Choose Table → Finish.

New Item - EntityFramework

Installed Sort by: Default Search (Ctrl+E)

Category	Item	Type	Description
Visual C#	ADO.NET Entity Data Model	Visual C#	Type: Visual C# A project item for creating an Entity Data Model.
	DataSet	Visual C#	
	EF 5.x DbContext Generator	Visual C#	
	EF 6.x DbContext Generator	Visual C#	
	SQL Server Database	Visual C#	
	XML File	Visual C#	
	XML Schema	Visual C#	
Online			

What should the model contain?

 EF Designer from database	 Empty EF Designer model	 Empty Code First model	 Code First from database
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Creates a model in the EF Designer based on an existing database. You can choose the data connection, settings for the model, and database objects to include in the model. The classes that your application will interact with are generated from the model.

Which database objects do you want to include in your model?

- ☒  **Tables**
 - ☒  **dbo**
 - ☒  **product**
 - ☐  Views
 - ☐  Stored Procedures and Functions

☒ Pluralize or singularize generated object names

☒ Include foreign key columns in the model

☐ Import selected stored procedures and functions into the entity model

Model Namespace:

Database1Model

< Previous Next > **Finish** Cancel

Step 5: UI Design

Webform.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="linq.WebForm1" %>

<!DOCTYPE html>

<html>
<head runat="server">
<title></title>
</head>
<body>
<form id="form1" runat="server">
<div>
<asp:Label Text="ID" runat="server" />
<asp:TextBox ID="txtId" runat="server" /><br /><br />
<asp:Label Text="Name" runat="server" />

```

```

<asp:TextBox ID="txtName" runat="server" /><br /><br />

<asp:Label Text="Age" runat="server" />
<asp:TextBox ID="txtAge" runat="server" /><br /><br />

<asp:Label Text="Div" runat="server" />
<asp:TextBox ID="txtDiv" runat="server" /><br /><br />

<asp:Label Text="Dream Job" runat="server" />
<asp:TextBox ID="txtJob" runat="server" /><br /><br />

<asp:Button Text="Insert" runat="server" OnClick="btnInsert_Click" />
<asp:Button Text="Update" runat="server" OnClick="btnUpdate_Click" />
<asp:Button Text="Delete" runat="server" OnClick="btnDelete_Click" />
<asp:Button Text="Select" runat="server" OnClick="btnSelect_Click" />

<br /><br />

<asp:GridView ID="GridView1" runat="server" />
</div>
</form>
</body>
</html>

```

Step 6: Backend Code

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace linq
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        Database2Entities dc = new Database2Entities();

        protected void Page_Load(object sender, EventArgs e)
        {

        }

        public void ShowData()
        {
            GridView1.DataSource = dc.Students.ToList();
            GridView1.DataBind();
        }

        protected void btnInsert_Click(object sender, EventArgs e)
        {
            Student s = new Student();

            s.Id = Convert.ToInt32(txtId.Text);
            s.Name = txtName.Text;

```

```

        s.Age = Convert.ToInt32(txtAge.Text);
        s.Div = txtDiv.Text;
        s.DreamJob = txtJob.Text;

        dc.Students.Add(s);
        dc.SaveChanges();

        Response.Write("Inserted Succesfully");
    }

    protected void btnUpdate_Click(object sender, EventArgs e)
    {
        int id = Convert.ToInt32(txtId.Text);

        Student s = dc.Students.Single(x => x.Id == id);

        s.Name = txtName.Text;
        s.Age = Convert.ToInt32(txtAge.Text);
        s.Div = txtDiv.Text;
        s.DreamJob = txtJob.Text;

        dc.SaveChanges();

        ShowData();
    }

    protected void btnDelete_Click(object sender, EventArgs e)
    {
        int id = Convert.ToInt32(txtId.Text);

        Student s = dc.Students.Single(x => x.Id == id);

        dc.Students.Remove(s);
        dc.SaveChanges();

        ShowData();
    }

    protected void btnSelect_Click(object sender, EventArgs e)
    {
        ShowData();
    }
}
}

```

ID

Name

Age

Div

Dream Job

Id	Name	Age	Div	DreamJob
1	Amna	25	B	Still Dreaming
2	ABC	22	B	Still Dreaming